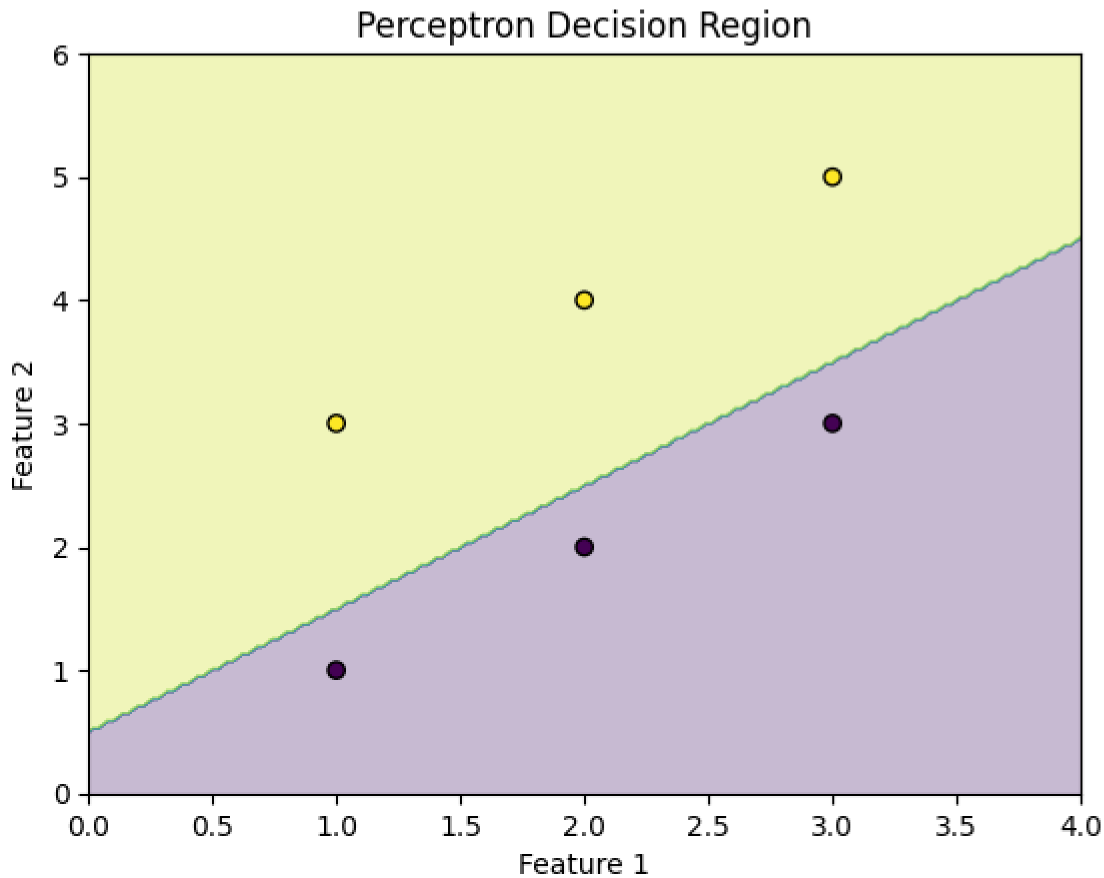
In [9]:

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import Perceptron

# Training data
X = np.array([[1,1], [2,2], [3,3], [1,3], [2,4], [3,5]])
y = np.array([0, 0, 0, 1, 1, 1])

# Train perceptron
model = Perceptron(max_iter=1000, eta0=0.1)
model.fit(X, y)    # ✅ Important

# Create decision region
xx, yy = np.meshgrid(
    np.linspace(0, 4, 200),
    np.linspace(0, 6, 200)
)

Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot
plt.contourf(xx, yy, Z, alpha=0.3)
plt.scatter(X[:,0], X[:,1], c=y, edgecolor='k')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Perceptron Decision Region")
plt.show()
```


Perceptron Decision Region
Feature 2
6
5
4
3
2
1
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Feature 1

```
In [ ]:
```